

Operating Systems - B.Tech Notes

CPU Scheduling:

CPU scheduling decides which process runs next on the processor.

Common algorithms: Round Robin, FCFS, SJF, Priority Scheduling.

Round Robin gives equal time slots called quantum to each process.

Deadlock:

Deadlock is a situation where two or more processes wait forever.

Each process holds a resource and waits for another held by others.

Four conditions for deadlock: Mutual Exclusion, Hold and Wait,

No Preemption, Circular Wait.

Database Management Systems - B.Tech Notes

Normalization:

Normalization organizes database to reduce redundancy.

1NF: Each column has atomic values, no repeating groups.

2NF: No partial dependency on primary key.

3NF: No transitive dependency.

Transactions:

A transaction is a unit of work that must complete fully or not at all.

ACID properties: Atomicity, Consistency, Isolation, Durability.